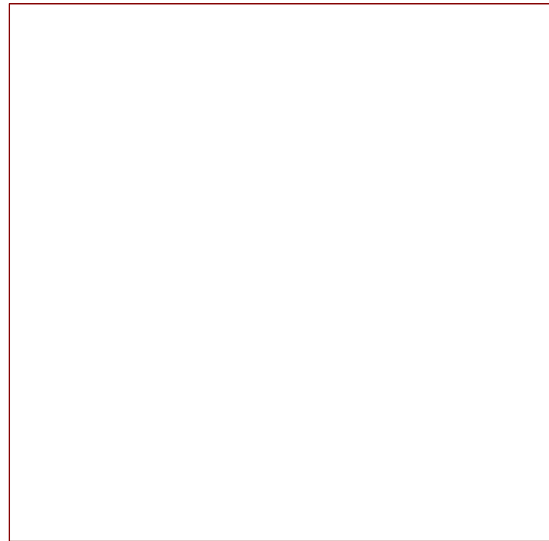




The diagram illustrates four ESC Modules arranged in a 2x2 grid. Each module is represented by a red square border. To the left of each module is a green arrow pointing to a motor label. The connections are as follows:

- ESC Module 1** is connected to **M1** (MOTOR).
- ESC Module 2** is connected to **M2** (MOTOR).
- ESC Module 3** is connected to **M3** (MOTOR).
- ESC Module 4** is connected to **M4** (MOTOR).

# Indicators

The diagram shows two parallel LED circuits. The top circuit consists of a green LED labeled D1 in series with a 1k resistor labeled R1, connected to a 5V supply labeled V\_3V3. The bottom circuit consists of a green LED labeled D2 in series with a 10k resistor labeled R3, connected to a 5V supply labeled V\_10V. Both LEDs are connected to ground (GND).

TVS

D3  
SMAJ20A

GND

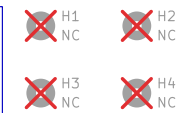
V\_BAT

The diagram shows a TVS diode labeled D3 with part number SMAJ20A. It is connected in parallel between the ground (GND) and the battery (V\_BAT) rails. The cathode of the diode is connected to GND, and the anode is connected to V\_BAT.

# Conns

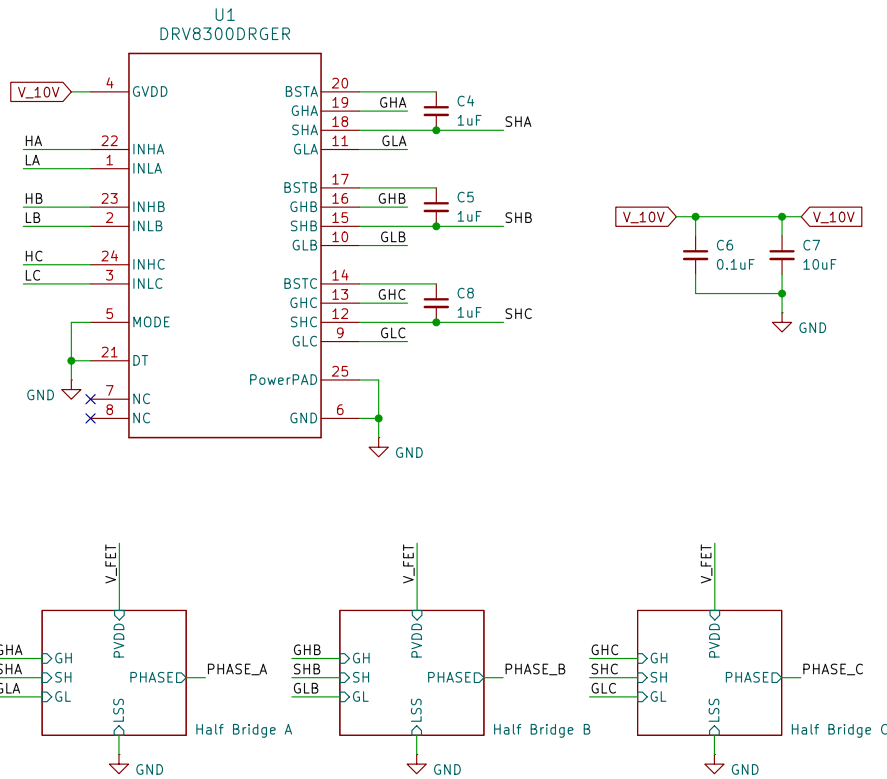
Diagram illustrating the connection scheme for the J1 connector (TO\_FC) and other components:

- J1 TO\_FC: Pins 1 through 9 are connected to M4, M3, M2, M1, and a common ground (GND) respectively.
- TELEM: Connected to pin 6 of J1.
- V\_BAT: Two modules connected to pins 7 and 8 of J1.
- J2 BAT\_IN: Pin 1 is connected to a V\_BAT module.
- J3 BAT\_GND: Pin 1 is connected to a common ground (GND).

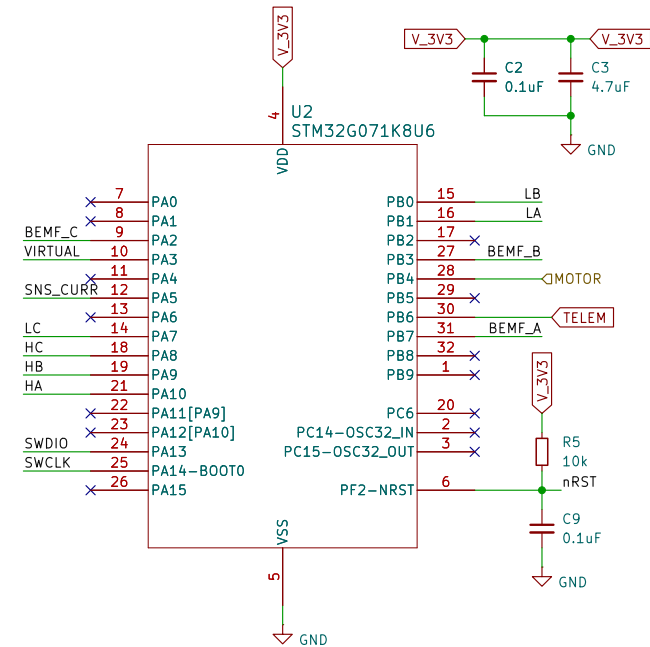


Id: 1/18

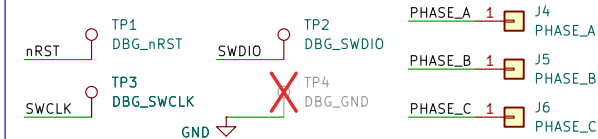
## Gate driver + half bridges



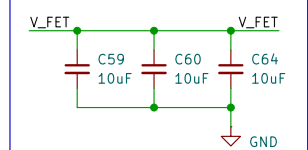
## MCU



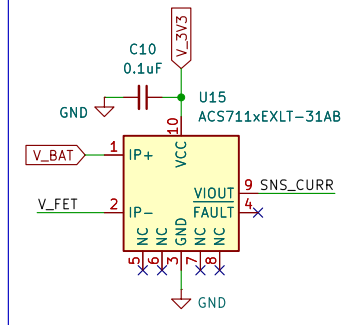
## Conns



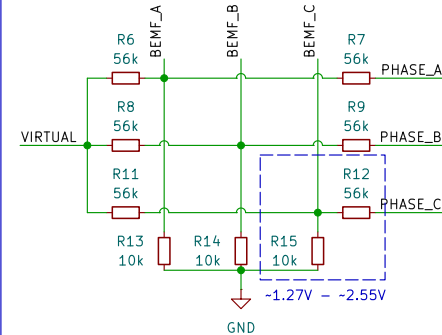
## MLCCs



## Current sensing



## Back-EMF sensing



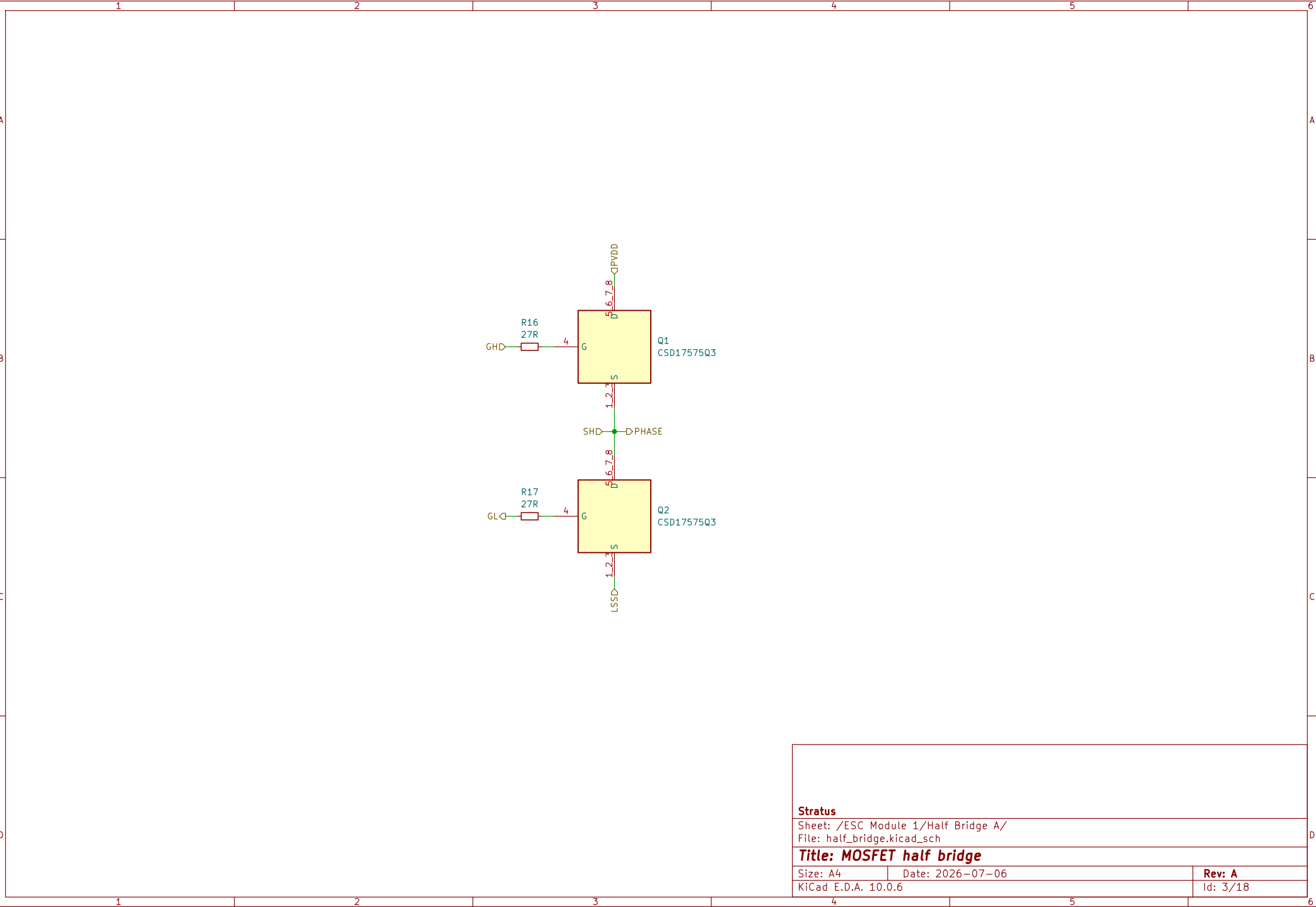
## Stratus

Sheet: /ESC Module 1/  
File: esc\_module.kicad\_sch

## Title: ESC Submodule

Size: A4 Date: 2026-07-06  
KiCad E.D.A. 10.0.6

Rev: A  
Id: 2/18

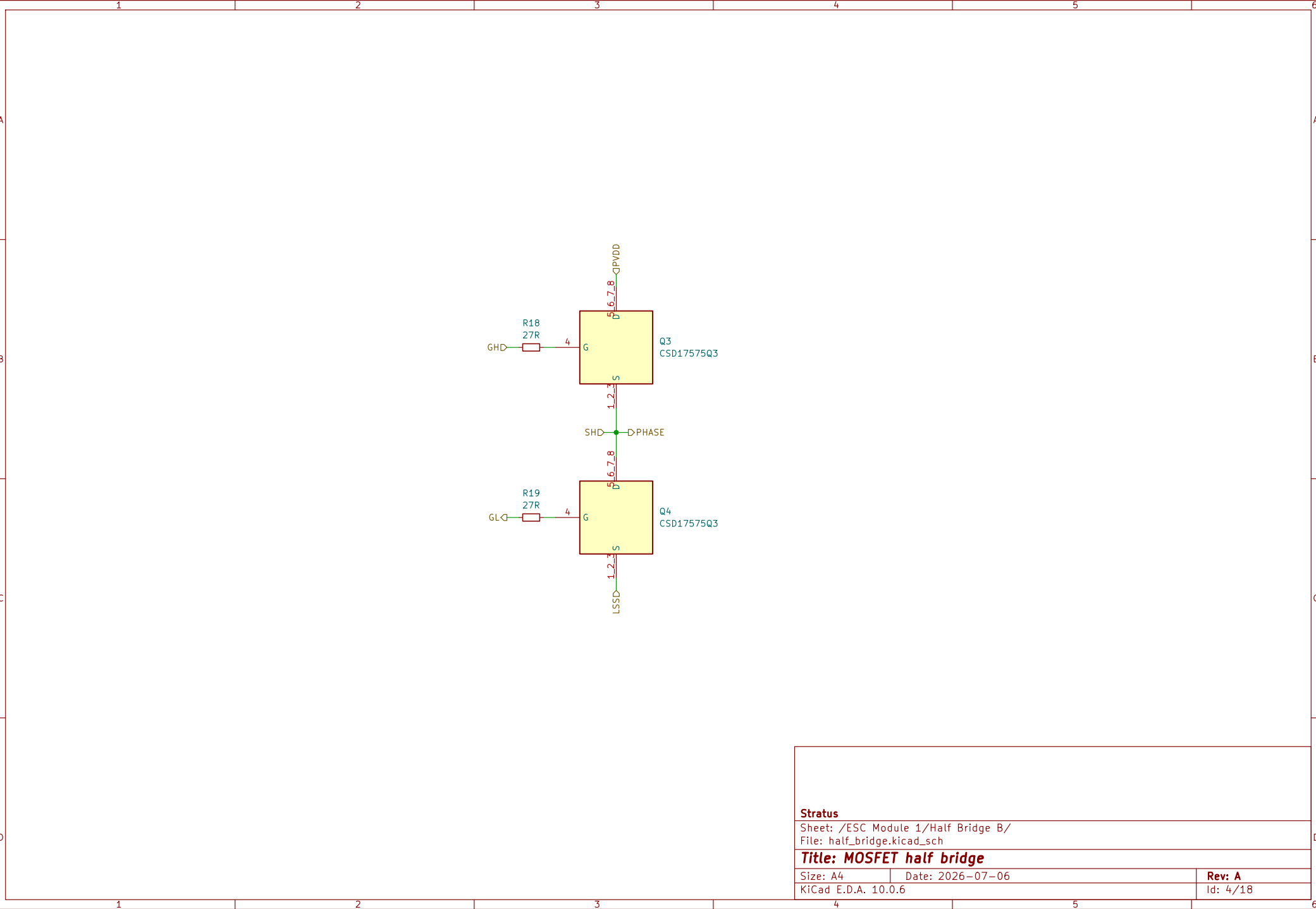


**Status**

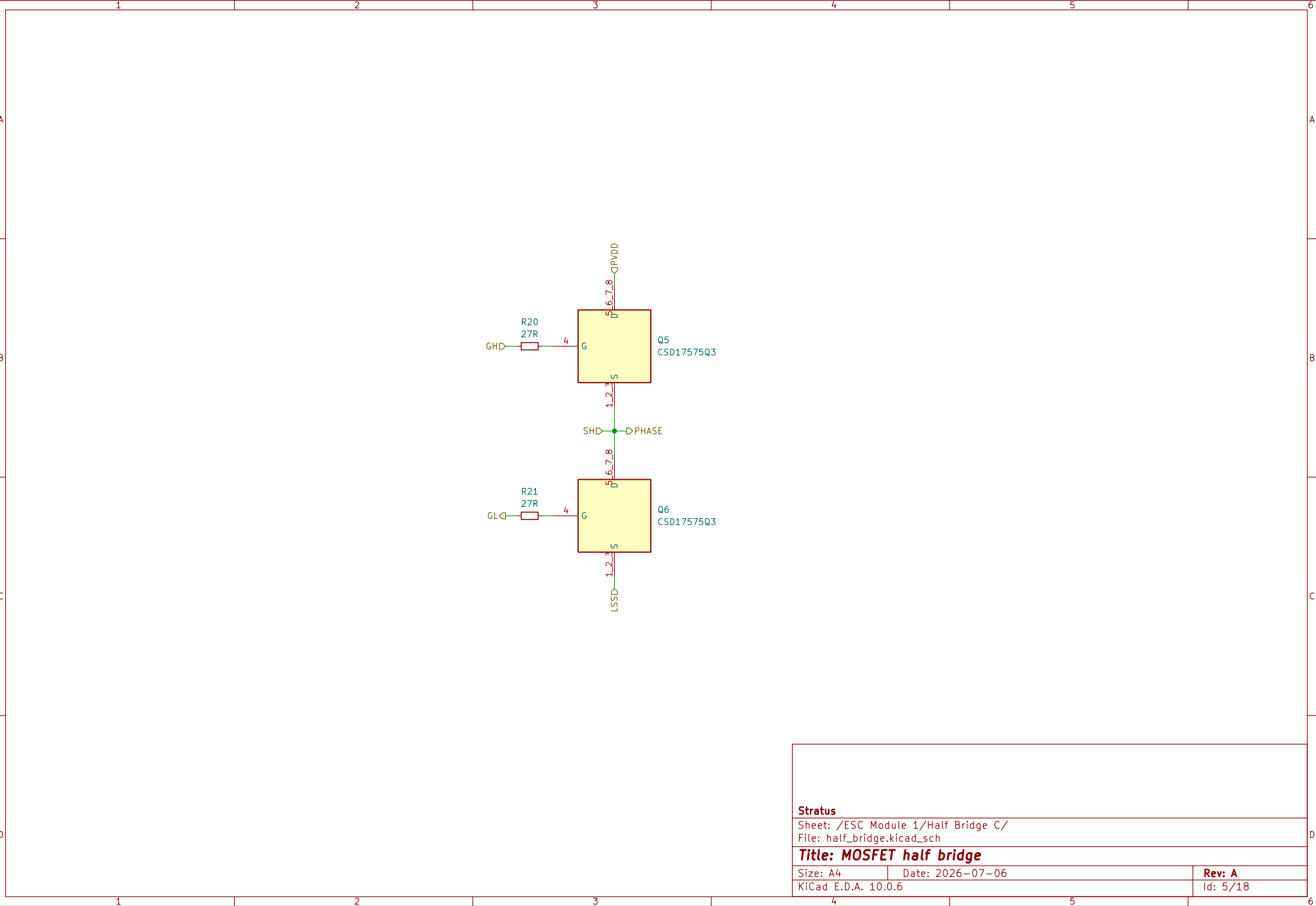
Sheet: /ESC Module 1/Half Bridge A/  
File: half\_bridge.kicad\_sch

**Title: MOSFET half bridge**

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6 |                  | Id: 3/18 |

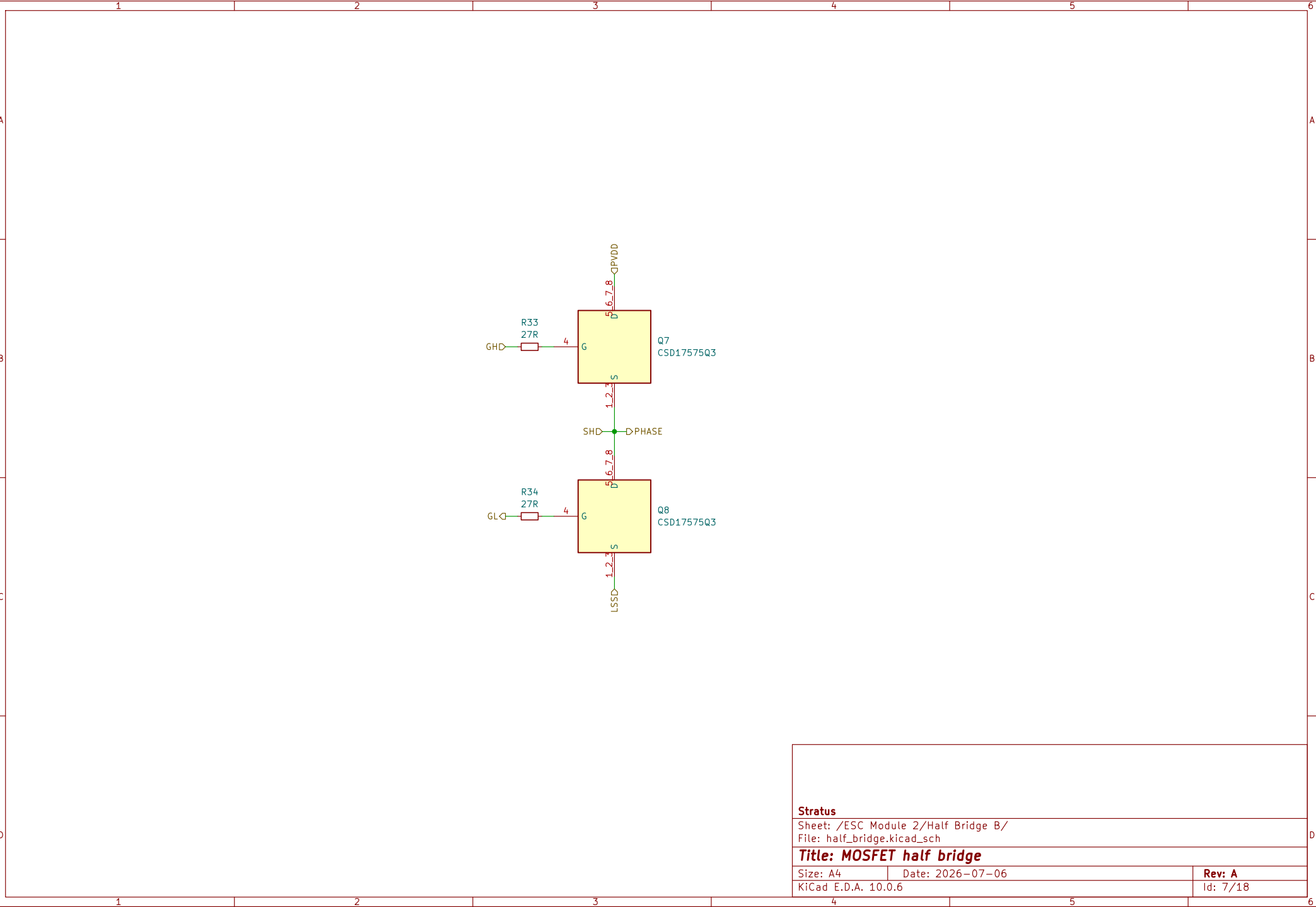


|                                     |                  |          |
|-------------------------------------|------------------|----------|
| Stratus                             |                  |          |
| Sheet: /ESC Module 1/Half Bridge B/ |                  |          |
| File: half_bridge.kicad_sch         |                  |          |
| Title: MOSFET half bridge           |                  |          |
| Size: A4                            | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6                 |                  | Id: 4/18 |



|                                                                    |                  |          |
|--------------------------------------------------------------------|------------------|----------|
| Stratus                                                            |                  |          |
| Sheet: /ESC Module 1/Half Bridge C/<br>File: half_bridge.kicad_sch |                  |          |
| Title: MOSFET half bridge                                          |                  |          |
| Size: A4                                                           | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6                                                |                  | Id: 5/18 |



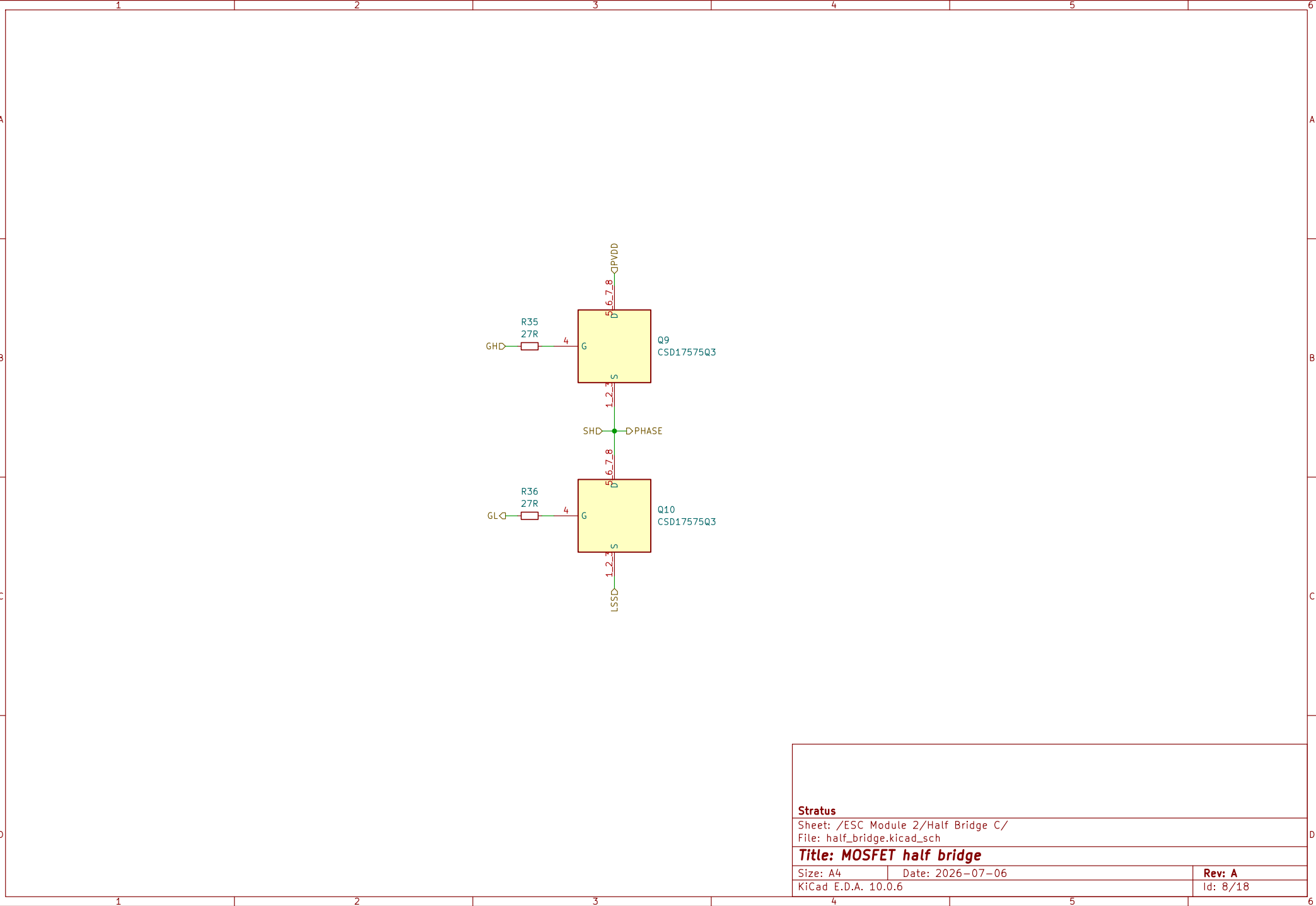


Stratus

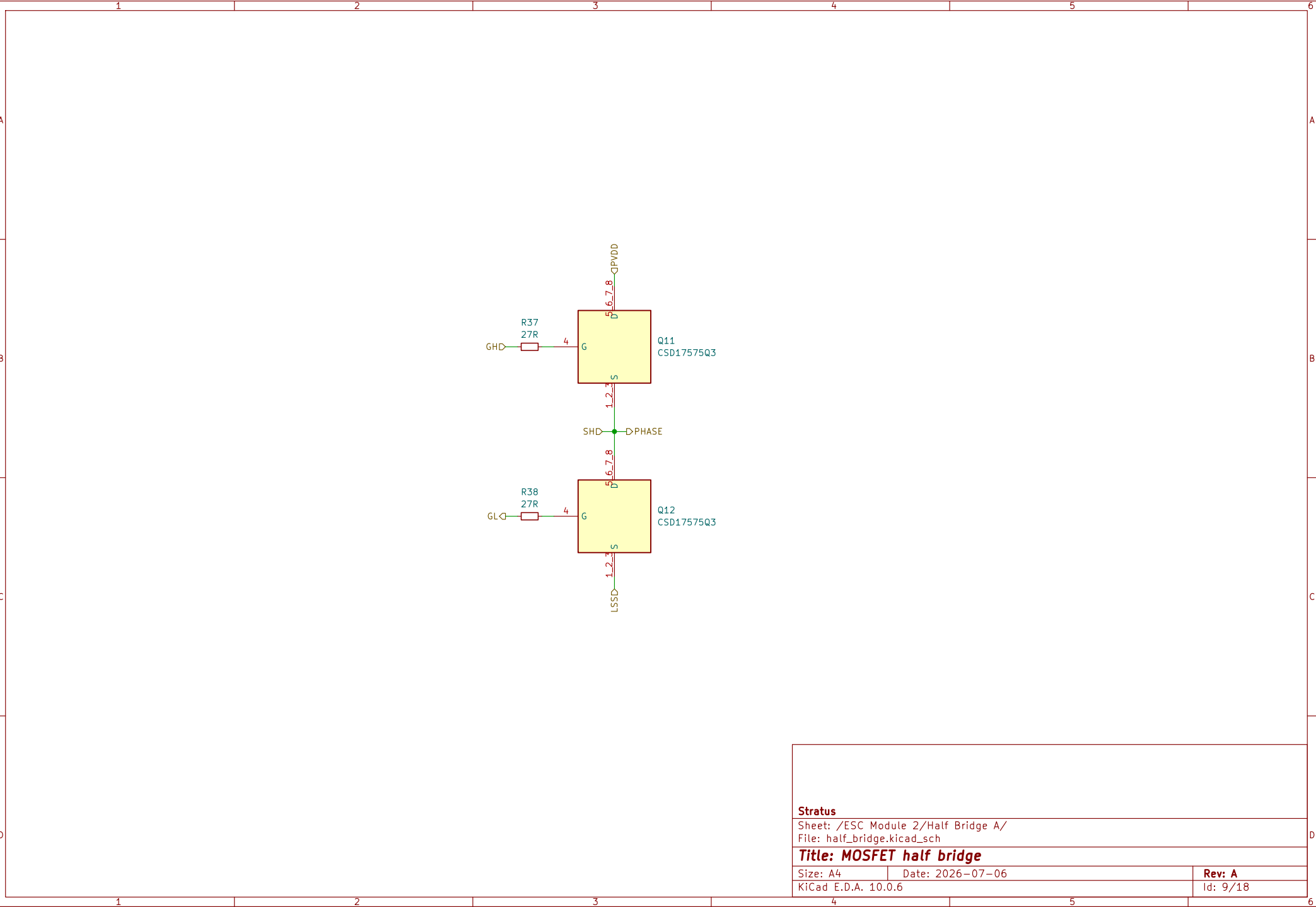
Sheet: /ESC Module 2/Half Bridge B/  
File: half\_bridge.kicad\_sch

Title: MOSFET half bridge

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6 |                  | Id: 7/18 |



|                                                                    |                  |          |
|--------------------------------------------------------------------|------------------|----------|
| Stratus                                                            |                  |          |
| Sheet: /ESC Module 2/Half Bridge C/<br>File: half_bridge.kicad_sch |                  |          |
| Title: MOSFET half bridge                                          |                  |          |
| Size: A4                                                           | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6                                                |                  | Id: 8/18 |



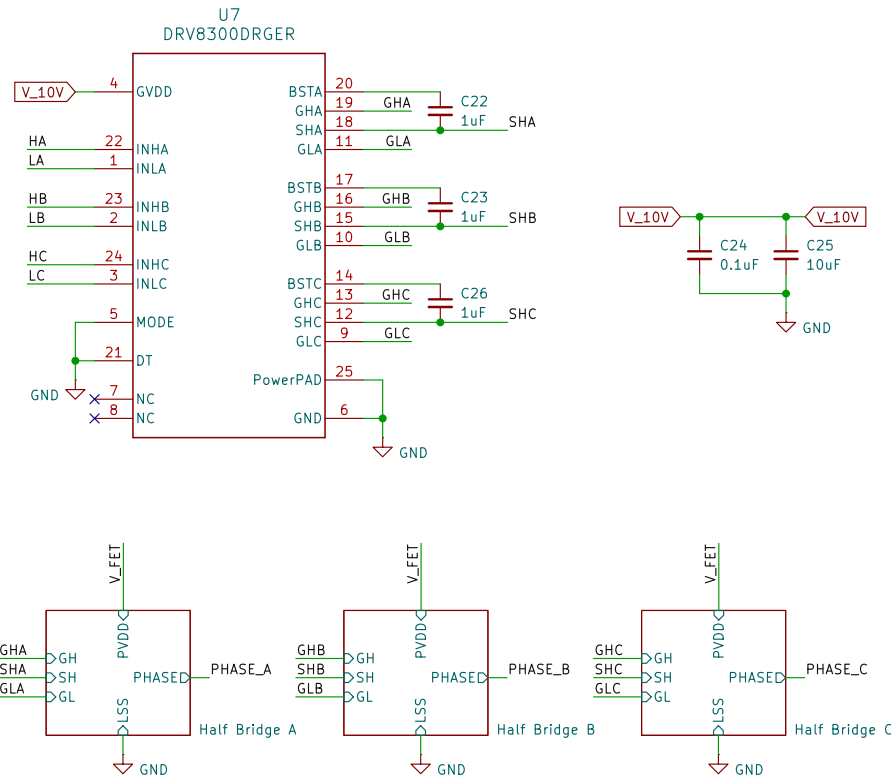
Stratus

Sheet: /ESC Module 2/Half Bridge A/  
File: half\_bridge.kicad\_sch

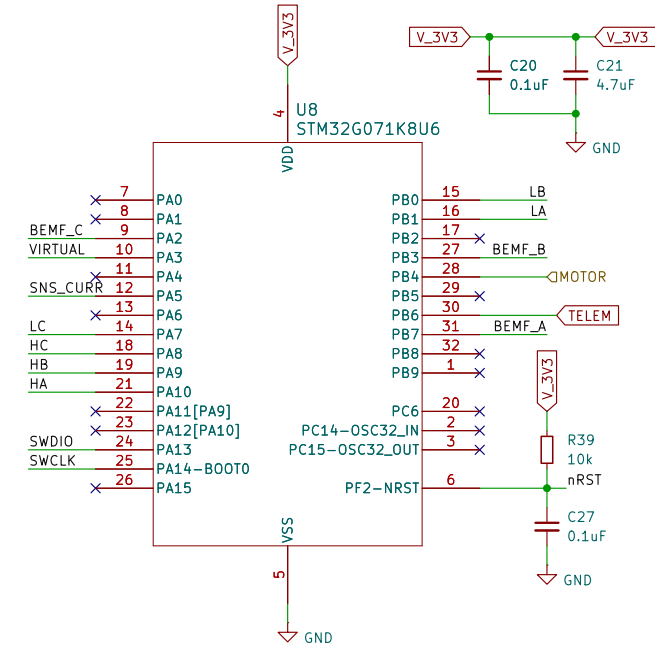
Title: MOSFET half bridge

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-07-06 | Rev: A   |
| KiCad E.D.A. 10.0.6 |                  | Id: 9/18 |

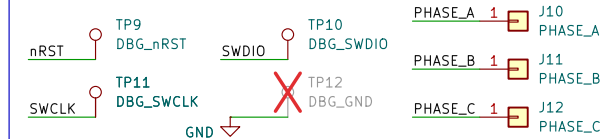
## Gate driver + half bridges



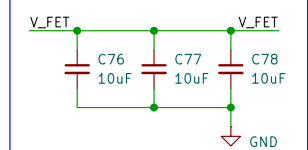
## MCU



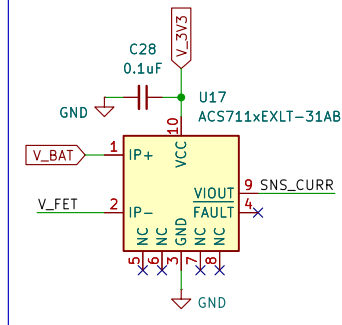
## Conns



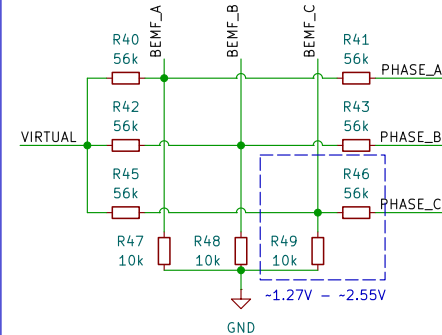
## MLCCs



## Current sensing



## Back-EMF sensing



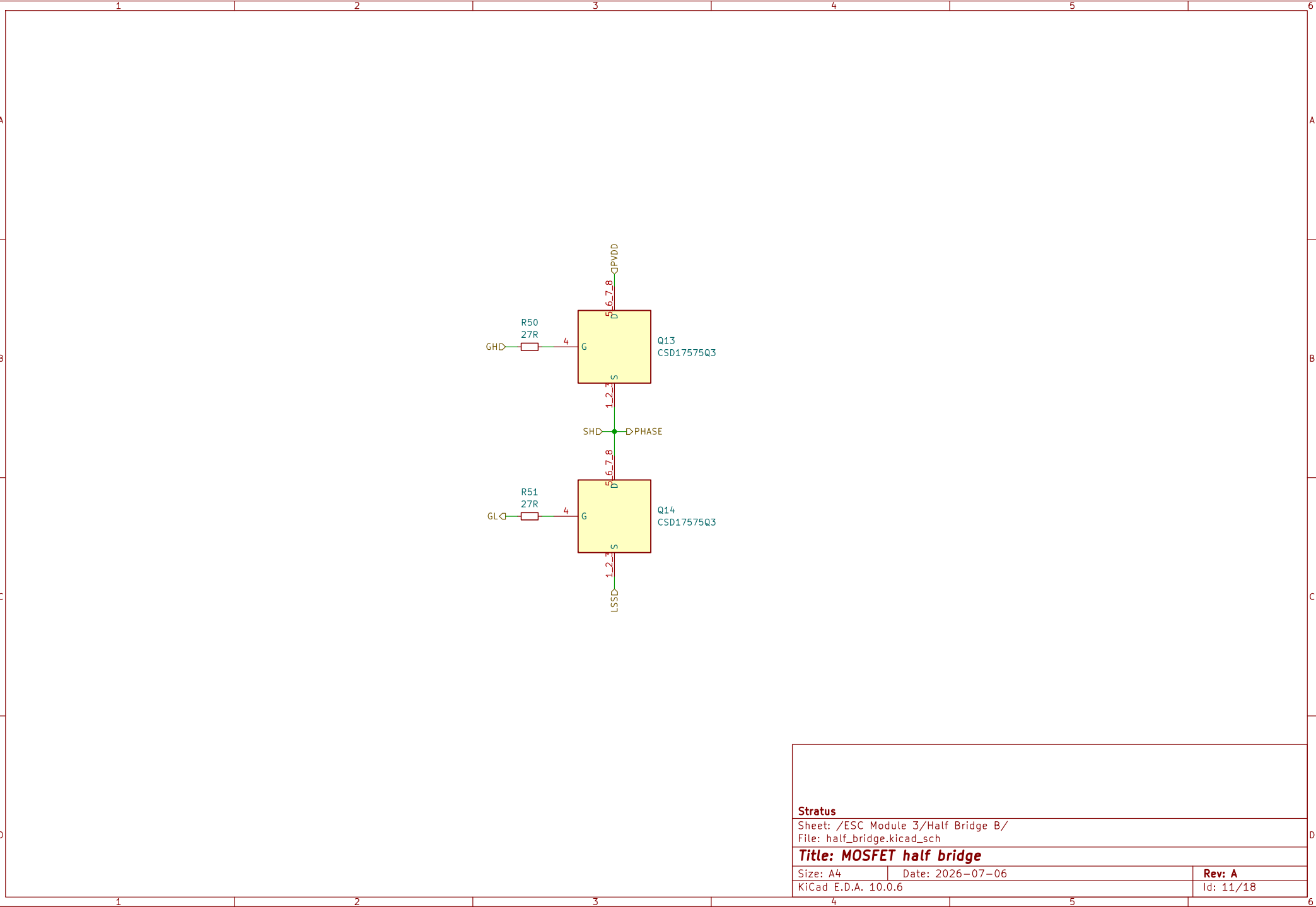
## Stratus

Sheet: /ESC Module 3/  
File: esc\_module.kicad\_sch

## Title: ESC Submodule

Size: A4 Date: 2026-07-06  
KiCad E.D.A. 10.0.6

Rev: A  
Id: 10/18

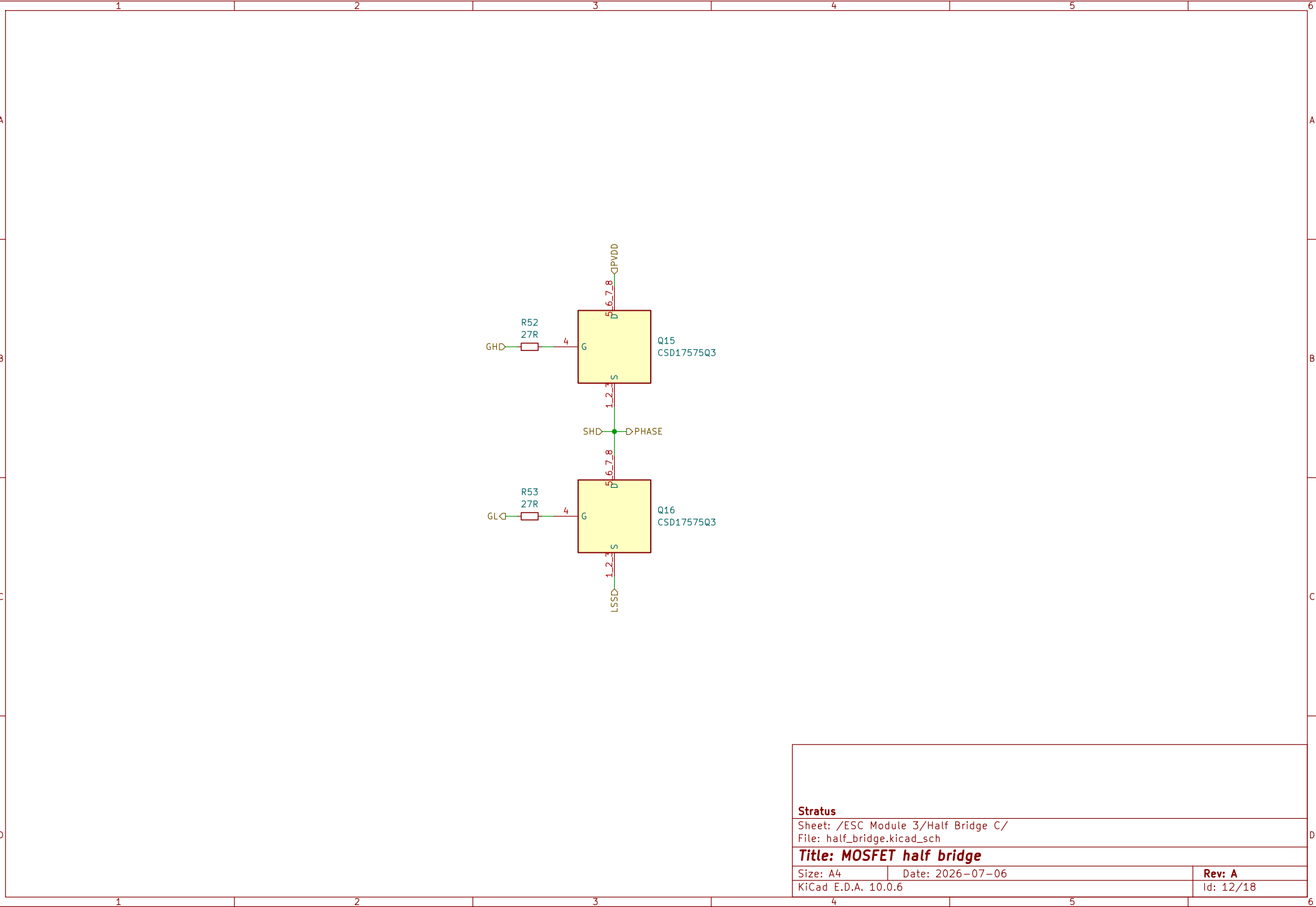


Stratus

Sheet: /ESC Module 3/Half Bridge B/  
File: half\_bridge.kicad\_sch

Title: MOSFET half bridge

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A4            | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6 |                  | Id: 11/18 |

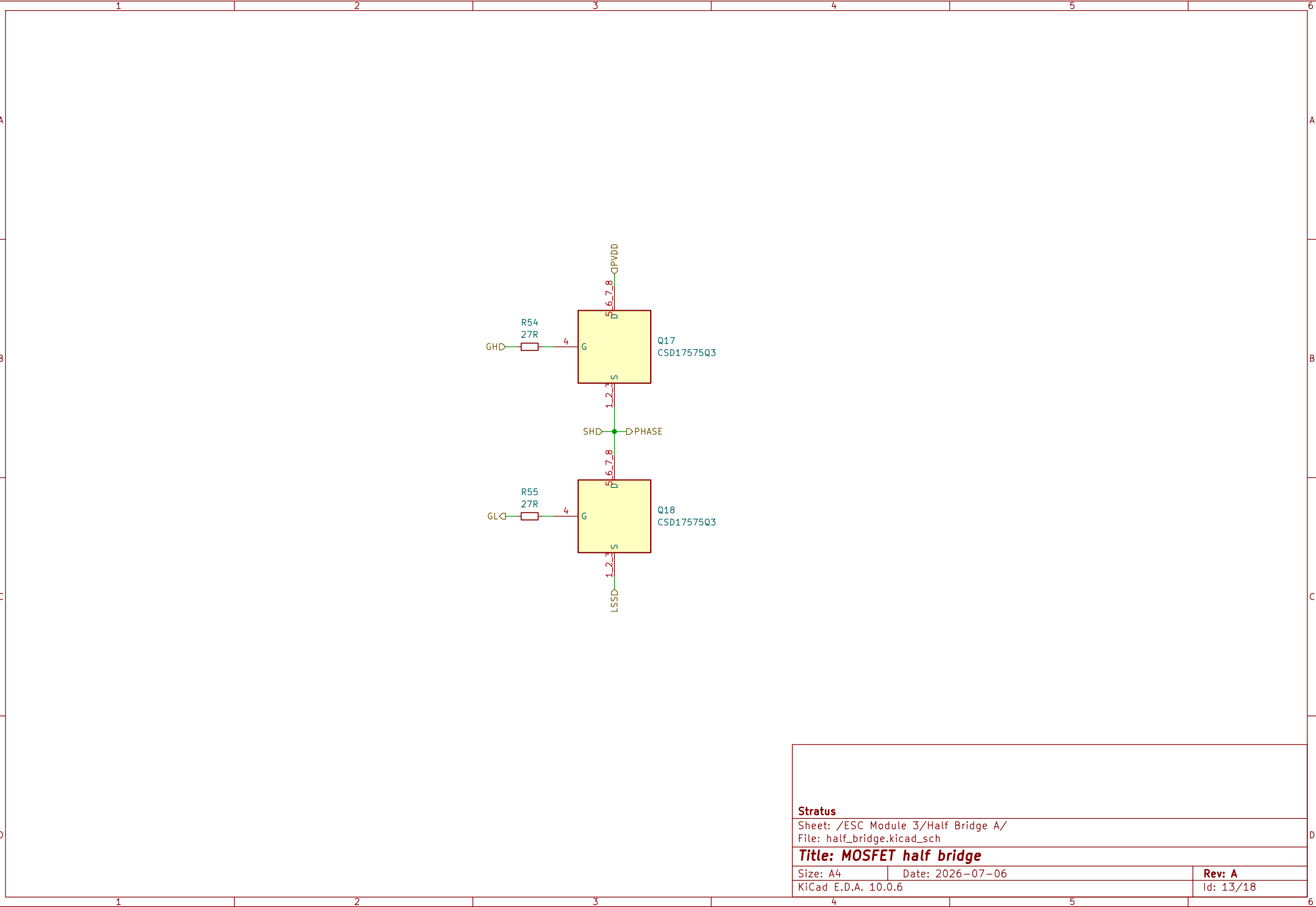


Stratus

Sheet: /ESC Module 3/Half Bridge C/  
File: half\_bridge.kicad\_sch

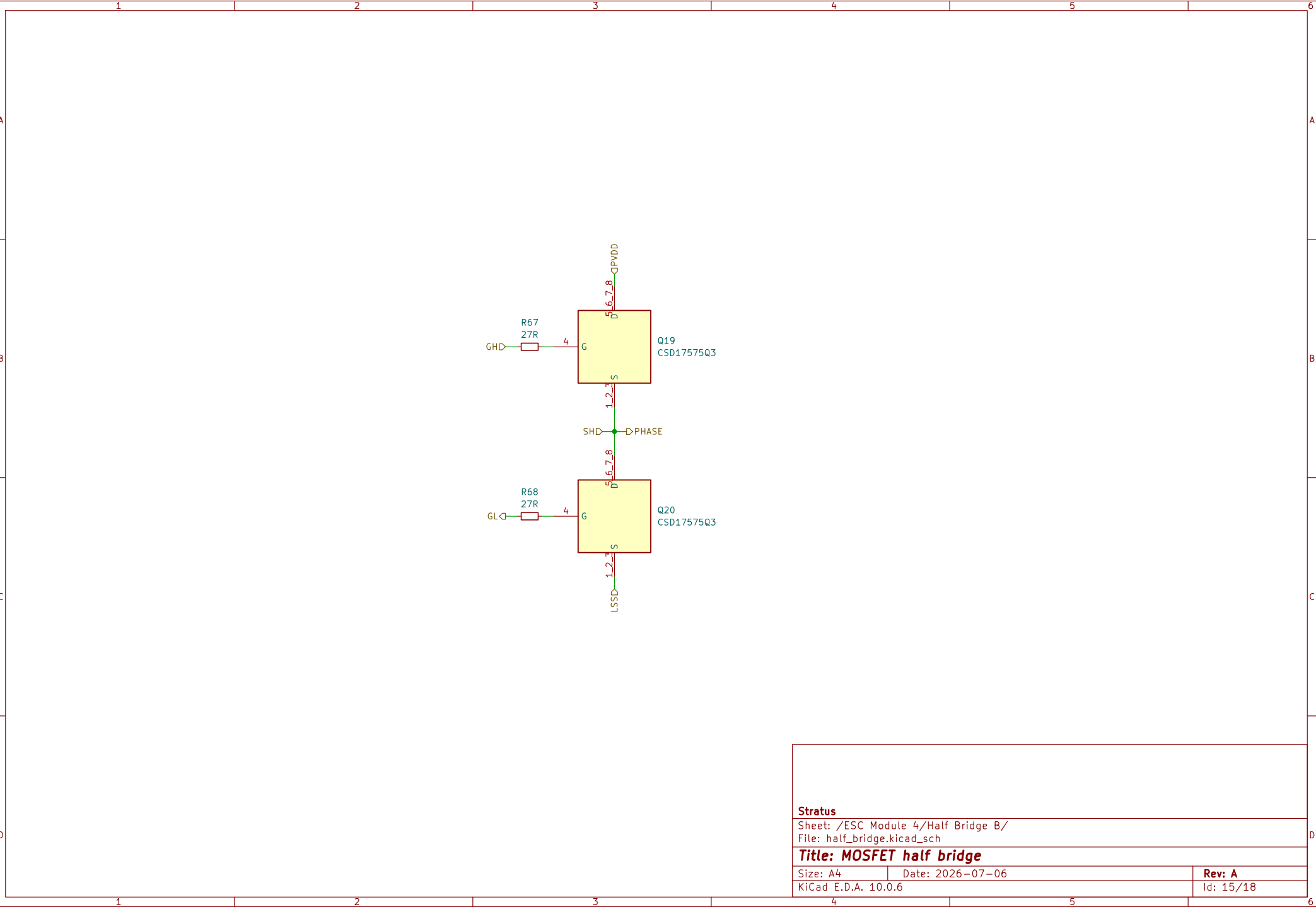
Title: MOSFET half bridge

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A4            | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6 |                  | Id: 12/18 |

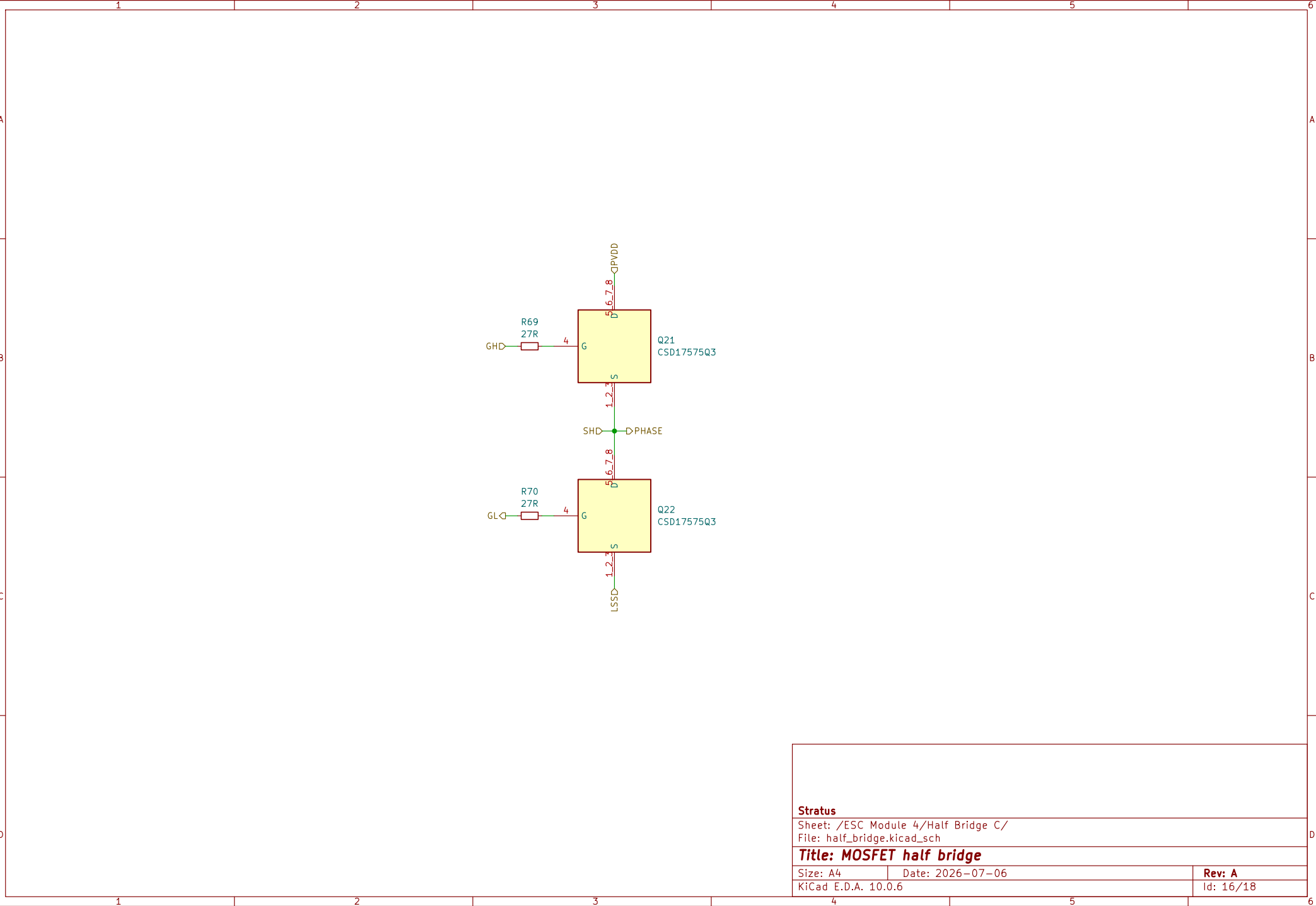


|                                                                    |                  |           |
|--------------------------------------------------------------------|------------------|-----------|
| Stratus                                                            |                  |           |
| Sheet: /ESC Module 3/Half Bridge A/<br>File: half_bridge.kicad_sch |                  |           |
| Title: MOSFET half bridge                                          |                  |           |
| Size: A4                                                           | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6                                                |                  | Id: 13/18 |

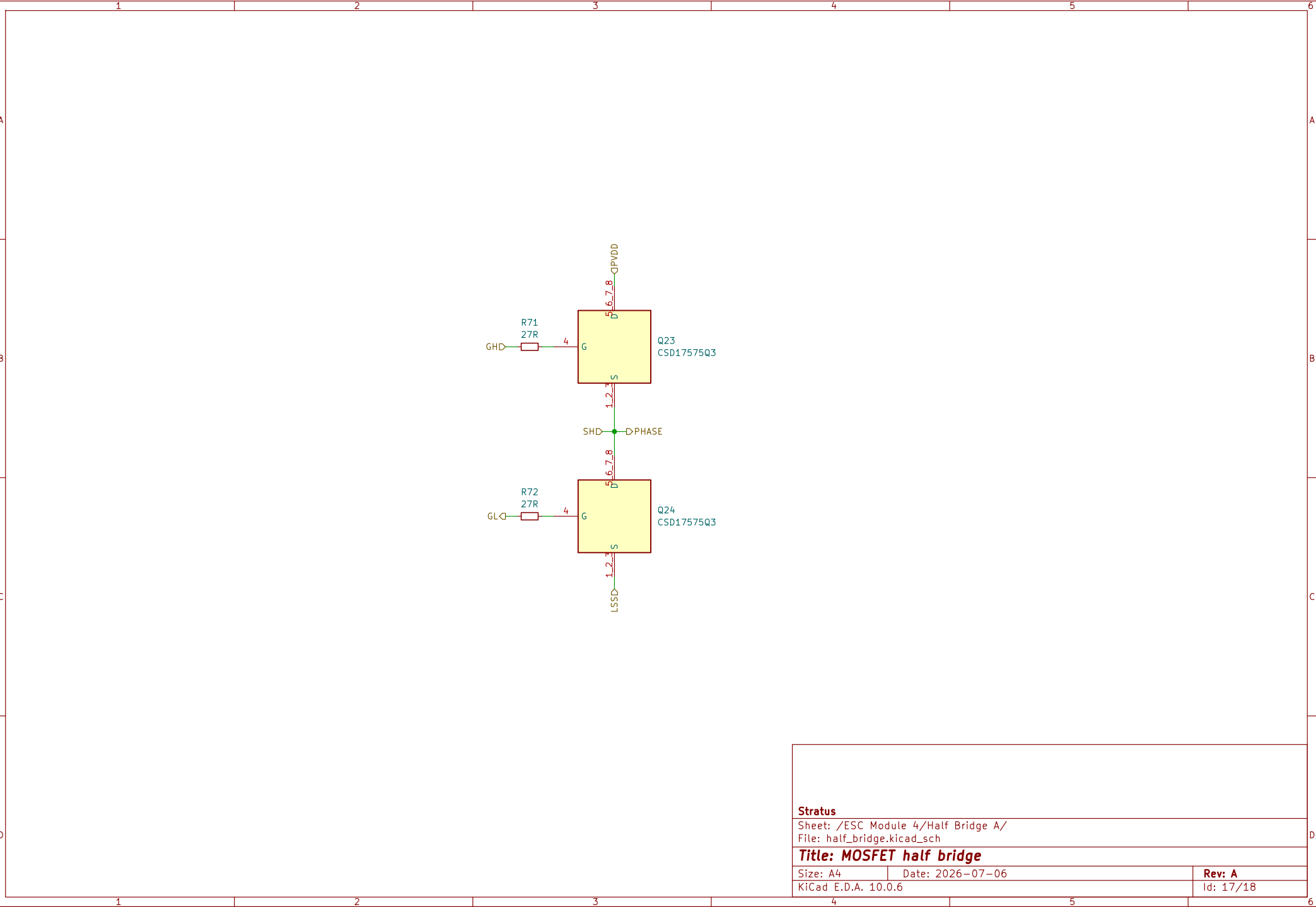




|                                                                    |                  |           |
|--------------------------------------------------------------------|------------------|-----------|
| Stratus                                                            |                  |           |
| Sheet: /ESC Module 4/Half Bridge B/<br>File: half_bridge.kicad_sch |                  |           |
| Title: MOSFET half bridge                                          |                  |           |
| Size: A4                                                           | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6                                                |                  | Id: 15/18 |



|                                     |                  |           |
|-------------------------------------|------------------|-----------|
| Stratus                             |                  |           |
| Sheet: /ESC Module 4/Half Bridge C/ |                  |           |
| File: half_bridge.kicad_sch         |                  |           |
| Title: MOSFET half bridge           |                  |           |
| Size: A4                            | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6                 |                  | Id: 16/18 |

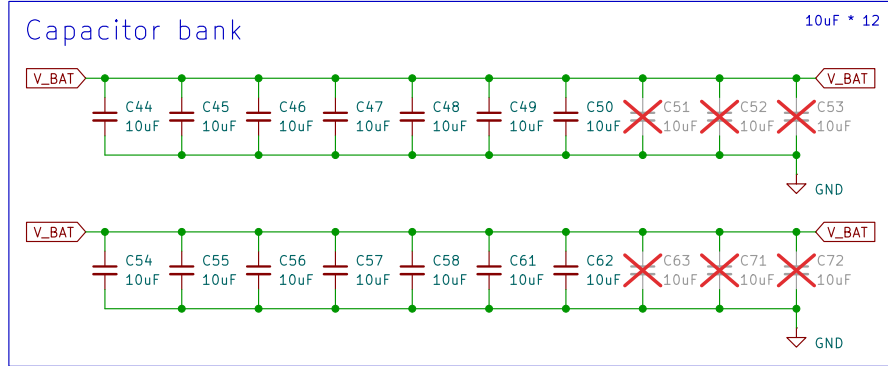
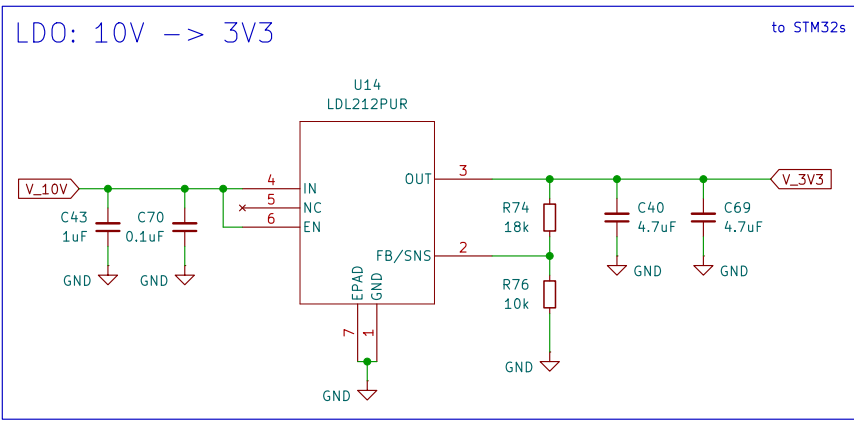
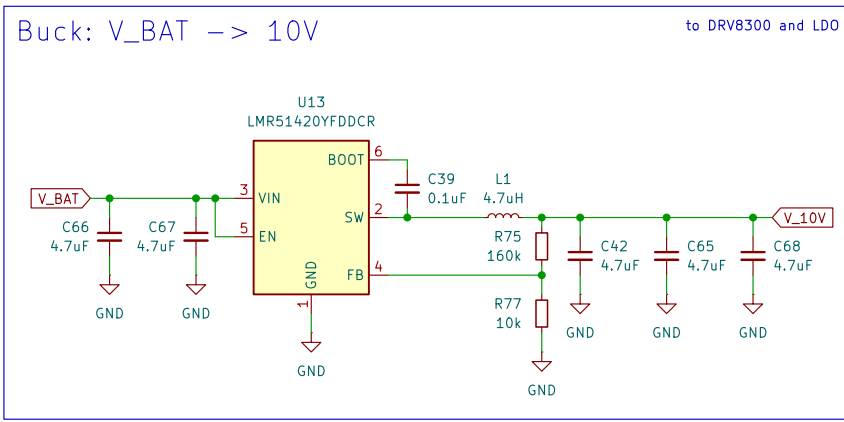


**Status**

Sheet: /ESC Module 4/Half Bridge A/  
File: half\_bridge.kicad\_sch

**Title: MOSFET half bridge**

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A4            | Date: 2026-07-06 | Rev: A    |
| KiCad E.D.A. 10.0.6 |                  | Id: 17/18 |



|                       |                  |           |
|-----------------------|------------------|-----------|
| Stratus               |                  |           |
| Sheet: /Power/        |                  |           |
| File: power.kicad_sch |                  |           |
| Title: Power Stages   |                  |           |
| Size: A4              | Date: 2026-09-10 | Rev: 1    |
| KiCad E.D.A. 10.0.6   |                  | Id: 18/18 |